

INTERNATIONAL STANDARD

**Coaxial communication cables –
Part 1-104: Electrical test methods – Test for the stability of the capacitance of
cable versus temperature**



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2015 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

IEC Catalogue - webstore.iec.ch/catalogue

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

IEC publications search - www.iec.ch/searchpub

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

Electropedia - www.electropedia.org

The world's leading online dictionary of electronic and electrical terms containing more than 30 000 terms and definitions in English and French, with equivalent terms in 15 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

IEC Glossary - std.iec.ch/glossary

More than 60 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

IEC Customer Service Centre - webstore.iec.ch/csc

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: csc@iec.ch.



IEC 61196-1-104

Edition 2.0 2015-08

INTERNATIONAL STANDARD

**Coaxial communication cables –
Part 1-104: Electrical test methods – Test for the stability of the capacitance of
cable versus temperature**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 33.120.10

ISBN 978-2-8322-2854-8

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

FOREWORD.....	3
1 Scope.....	5
2 Normative references.....	5
3 Terms and definitions	5
4 Test method	5
4.1 Equipment	5
4.2 Test sample	5
4.3 Procedure	5
5 Expression of test results.....	6
6 Test report.....	6
7 Requirements	6
Table 1 – Temperature cycling	6

INTERNATIONAL ELECTROTECHNICAL COMMISSION

COAXIAL COMMUNICATION CABLES –**Part 1-104: Electrical test methods –
Test for the stability of the capacitance of cable versus temperature**

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as “IEC Publication(s)”). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 61196-1-104 has been prepared by subcommittee 46A: Coaxial cables, of IEC technical committee 46: Cables, wires, waveguides, R.F. connectors, R.F. and microwave passive components and accessories.

This second edition cancels and replaces the first edition published in 2005. This edition constitutes a technical revision. This edition includes the following significant technical changes with respect to the previous edition:

- a) Subclause 4.2, Requirements for the test sample;
- b) Change in the title of Part 1-104.

The text of this standard is based on the following documents:

FDIS	Report on voting
46A/1263A/FDIS	46A/1268/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 61196 series published under the general title *Coaxial communication cables* can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

COAXIAL COMMUNICATION CABLES –

Part 1-104: Electrical test methods – Test for the stability of the capacitance of cable versus temperature

1 Scope

This part of IEC 61196 applies to coaxial communications cables. It specifies test methods for determining the capacitance stability of the cable when subjected to temperature change.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050 (all parts), *International Electrotechnical Vocabulary*, available at <http://www.electropedia.org/>

IEC 61196-1, *Coaxial communication cables – Part 1: Generic specification – General, definitions and requirements*

IEC 61196-1-103, *Coaxial communication cables – Part 1-103: Electrical test methods – Test for capacitance of cable*

3 Terms and definitions

For the purpose of this document the terms and definitions given in IEC 60050 and in IEC 61196-1 apply.

4 Test method

4.1 Equipment

An environmental chamber of size and dimension capable of performing the test described herein is required.

The capacitance equipment shall be in accordance with IEC 61196-1-103.

4.2 Test sample

The test sample shall be in accordance with IEC 61196-1-103

4.3 Procedure

The capacitance shall be measured in accordance with IEC 61196-1-103.

The cable under test (CUT) shall be subjected to the applicable temperature cycle shown in Table 1 for a total of three cycles unless otherwise stated in the relevant sectional or detail specification.

The capacitance of the CUT shall be measured initially and after each step at the actual test temperature.

Table 1 – Temperature cycling

Step	Foamed PE dielectric type Temperature °C	PE dielectric type Temperature °C	PTFE dielectric type Temperature °C	Time hours
initial	20 ± 2	20 ± 2	20 ± 2	2 minimum
1	+65 ± 2	+75 ± 2	+250 ± 5	2 minimum
2	+20 ± 2	+20 ± 2	+20 ± 2	2 minimum
3	-40 ± 2	-40 ± 2	-55 ± 2	2 minimum
4	+20 ± 2	+20 ± 2	+20 ± 2	2 minimum

For other dielectric materials the test temperatures and testing hours shall be stated in the relevant sectional or detail specification.

5 Expression of test results

The test results should be normalized to the reference length of 1 m.

$$C = \frac{C_m}{L} (\text{pF/m})$$

where

C is the capacitance of reference length at measuring temperature;

C_m is the measured capacitance value of the CUT in picofarads;

L is the length of sample in metres.

6 Test report

The test report shall record the test conditions given below at each temperature step:

- temperatures,
- number of cycles, if different from three,
- sample length,
- test frequency,
- cable capacitance at each step

and record the deviation from the initial value in pF/m for each temperature step as stated in the relevant sectional or detail specification.

7 Requirements

The deviation of the capacitance from the initial value in pF/m for each temperature step of the CUT shall comply with the requirements of the relevant detailed specification.

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch